4339 Asset Pricing and Investments - Assignment 3

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1 Factors and Arbitrage

We assume that stock returns follow a factor structure with two factors f_1 and f_2. Furthermore, the risk-free rate $r_f = 3\%$, and A and B are two well-diversified portfolios with the following properties:

Portfolio	β_1	β_2	$E[r]$
A	1	0.5	7.5%
B	0.2	2	13.4%

1.1 (a) The price of risk for each factor, f_1 and f_2, under no arbitrage:

For a well-diversified portfolio idiosyncratic risk is diversified away, so expected excess returns are fully explained by the factor exposure,

$$E[r_i] - r_f = \lambda_1 \beta_{i1} + \lambda_2 \beta_{i2},$$

where λ_n is the excess return per unit of exposure to factor n.

Substituting from the initial table yields the following two equations:

$$\begin{aligned} A: &\quad 0.075 - 0.03 = \lambda_1 \cdot 1 + \lambda_2 \cdot 0.5 \\ B: &\quad 0.134 - 0.03 = \lambda_1 \cdot 0.2 + \lambda_2 \cdot 2 \end{aligned}$$

We find the price of risk for each factor by solving equation A for λ_1 and substituting into B:

$$\begin{aligned} A: &\quad \lambda_1 = 0.045 - 0.5\lambda_2 \\ B: &\quad 0.104 = (0.045 - 0.5\lambda_2) \cdot 0.2 + \lambda_2 \cdot 2 \\ &\quad 0.104 = 0.009 + 1.9\lambda_2 \\ &\quad \lambda_2 = 0.05 \quad \Longrightarrow \quad \lambda_1 = 0.02 \end{aligned}$$

The price of risk for factor f_1 is $\lambda_1 = 2.0\%$ and for factor f_2 it is $\lambda_2 = 5.0\%$, in this system.

1.2 (b) Identifying and exploiting an arbitrage opportunity:

C is a well-diversified portfolio with factor loading $(\beta_{C1}, \beta_{C2}) = (0.5,\ 1.0)$ and expected return $E^{\text{obs}}[r_C] = 9.5\%$:

Portfolio	β_{C1}	β_{C2}	$E^{\text{obs}}[r]$
C	0.5	1.0	9.5%

According to C's factor loadings and the prices of risk calculated in (a), the expected no-arbitrage return of C is

$$\begin{aligned} E^{\text{model}}[r_C] &= 0.03 + \lambda_1 \cdot 0.5 + \lambda_2 \cdot 1 \\ &= 9.0\% \end{aligned}$$

Because the value does not coincide with the observed return in the table, there exists an arbitrage opportunity.

To take advantage of it, we create a portfolio H made up of A, B and the risk-free asset, which replicates the factor exposure of C:

$$\begin{aligned} & 1 = w_A + w_B + w_f \\ f_1: \quad & 0.5 = w_A + 0.2\, w_B + 0\, w_f \\ f_2: \quad & 1 = 0.5\, w_A + 2\, w_B + 0\, w_f \end{aligned}$$

Solving equation f_1 for the weight in portfolio B and substituting into equation f_2 yields the respective weights in portfolio H:

$$\begin{aligned} f_1: \quad & w_B = 2.5 - 5w_A \\ f_2: \quad & 1 = 0.5w_A + 2 \cdot (2.5 - 5w_A) \\ & 4 = 9.5\, w_A \\ & w_A = \tfrac{16}{38} \quad \Longrightarrow \quad w_B = \tfrac{15}{38} \quad \Longrightarrow \quad w_f = \tfrac{7}{38} \end{aligned}$$

The return of this newly created portfolio H is

$$\begin{aligned} E^{\text{obs}}[r_H] &= \sum_{i=1}^{n} r_i^{\text{obs}} \cdot w_i \\ E^{\text{obs}}[r_H] &= 9.0\% \end{aligned}$$

Therefore, we would take a long position in portfolio C and a short position of 1 in H. The net investment is zero, the net exposure to each factor is zero, and as such the position earns a riskless profit of $9.5\% - 9.0\% = 0.5\%$ per unit of capital.

2 Performance measures

Consider the following properties for two portfolios:

Portfolio	$E[r^e]$	σ
Market (M)	10%	25%
P	15%	35%

The Sharpe ratios of the two portfolios are

$$S_M = \frac{E[r_M^e]}{\sigma_M} = \frac{0.10}{0.25} = 0.4000, \qquad S_P = \frac{E[r_P^e]}{\sigma_P} = \frac{0.15}{0.35} \approx 0.4286.$$

2.1 (a) Constructing P^*

In the $(\sigma, E[r])$ grid, all weight alllocations between a risky portfolio and the risk-free asset lie on a linear function through $(0, r_f)$, with the slope representing that portfolio's Sharpe ratio. By moving along this linear function, we can adjust $(\sigma, E[r])$, while the Sharpe ratio of P remains constant. In example, σ_P needs to be lowered to match σ_M. We therefore move left, since $\sigma_M < \sigma_P$.

Let the investor hold a weight w_P in portfolio P and $w_f = 1 - w_P$ in the risk-free asset. The excess return of the combined portfolio is

$$\begin{aligned} E[r_{P^*}] &= w_P\, r_P + (1 - w_P)\, r_f \\ E[r_{P^*}^e] &= w_P\, E[r_P^e] \end{aligned} \tag{1}$$

since the risk-free asset contributes no excess return. The portfolio volatility is

$$\begin{aligned} \sigma_{P^*} &= \sqrt{w_P^2\sigma_P^2 + w_f^2\sigma_f^2 + 2w_P w_f \sigma_{P,f}} \\ \sigma_{P^*} &= \sqrt{w_P^2\sigma_P^2} = w_P\, \sigma_P, \end{aligned}$$

since the risk-free asset has zero variance and zero covariance with P.

We see that both $E[r_{P^*}^e]$ and σ_{P^*} solely relate to the original levels of P scaled by the same factor w_P. This is required to ensure that the Sharpe ratio is unaffected by the weight distribution between P and the risk free asset, and formalize the initial "linear function" understanding.

By imposing the matching volatility constraint we can derive the required weight in portfolio P:

$$\sigma_{P^*} = w_P\, \sigma_P = \sigma_M w_P = \frac{\sigma_M}{\sigma_P} = \frac{0.25}{0.35} = \frac{5}{7} \approx 0.7143 \qquad \longrightarrow \qquad w_f = 1 - w_P = \frac{2}{7} \approx 0.2857$$

Substituting into (1) yields the expected excess return for P^*:

$$E[r_{P^*}^e] = w_P\, E[r_P^e] = \tfrac{5}{7} \cdot 15\% \approx 10.71\%$$

The portfolio P^* that matches the σ_M therefore invests 71.4% in P and 28.6% in the risk-free asset.

2.2 (b) Equivalence of the two comparisons

Following the results from (a), we know that P^* was derived in such a way that its Sharpe ratio matches P's:

$$S_{P^*} = \frac{E[r^e_{P^*}]}{\sigma_{P^*}} = \frac{w_P\, E[r^e_P]}{w_P\, \sigma_P} = \frac{E[r^e_P]}{\sigma_P} = S_P$$

The Sharpe ratio independent of the allocation between P and the risk-free asset as the scaling term cancels out.

Using this fact, $S_P = S_{P^*}$, and the initial constraint $\sigma_{P^*} = \sigma_M$, both terms can be rewritten to have a common denominator:

$$S_P - S_M = S_{P^*} - S_M = \frac{E[r^e_{P^*}]}{\sigma_{P^*}} - \frac{E[r^e_M]}{\sigma_M} = \frac{E[r^e_{P^*}] - E[r^e_M]}{\sigma_M}$$

The two comparisons therefore rank P and M identically. This can also be numerically verified as the comparison yields

$$S_P - S_M = 0.4286 - 0.4000 = 0.0286, \qquad \frac{E[r^e_{P^*}] - E[r^e_M]}{\sigma_M} = \frac{10.71\% - 10.00\%}{25\%} = 0.0286.$$

The second comparison could be more intuitive, because $E[r^e_{P^*}] - E[r^e_M] = 0.71$ percentage points answers how much additional expected return does the investor earn by choosing P over the market portfolio, while holding risk fixed at the market's level of 25%.

On the other hand, the Sharpe ratio comparison carries the advantage of being able to applied directly to the two portfolios, without the need to create a market-risk matched portfolio \$P^{*}.

2.3 (c) Optimal combination of P and M

Portfolio P offers a higher Sharpe ratio than the market, and a market beta of 1.2. Adhering to a single-factor we can seperate P's excess return in alpha, market exposure, and residual risk with no expected excess return:

$$r^e_P = \alpha_P + \beta_P\, r^e_M + \varepsilon_P, \qquad E[\varepsilon_P] = 0, \qquad \mathrm{Cov}(\varepsilon_P, r^e_M) = 0$$

The alpha of P is the expected excess return not explained by its market exposure:

$$\alpha_P = E[r^e_P] - \beta_P\, E[r^e_M] = 0.15 - 1.2 \cdot 0.10 = 0.03 = 3\%$$

The total portfolio varience consists of systematic, weighted by the market beta and idiosyncratic component. Solving for the idiosyncratic part yields:

$$\sigma_\varepsilon^2 = \sigma_P^2 - \beta_P^2 \sigma_M^2 = 0.35^2 - 1.2^2 \cdot 0.25^2 = 0.1225 - 0.0900 = 0.0325,$$

$$\sigma_\varepsilon = \sqrt{0.0325} \approx 0.1803 = 18.03\%, \qquad \mathrm{Cov}(r_P, r_M) = \beta_P \sigma_M^2 = 1.2 \cdot 0.0625 = 0.075.$$

To find the optimal weight w_P in portfolio P we compare its alpha per unit of residual variance with the market's reward per unit of market variance:

$$w_P^0 = \frac{\alpha_P/\sigma_\varepsilon^2}{E[r_M^e]/\sigma_M^2} = \frac{0.03/0.0325}{0.10/0.0625} = \frac{0.9231}{1.6000} = \frac{15}{26} \approx 0.5769.$$

Because P is not beta-neutral, a position in P also carries market exposure. We therefore adjust the weight by the portfolio's market beta

$$w_P = \frac{w_P^0}{1 + (1 - \beta_P)\, w_P^0} = \frac{15/26}{1 - 0.2 \cdot 15/26} = \frac{15/26}{23/26} = \frac{15}{23} \approx 0.6522$$

$$w_M = 1 - w_P = \frac{8}{23} \approx 0.3478.$$

The optimal risky portfolio therefore holds 65.2% in P and 34.8% in the market portfolio. Since $\beta_P > 1$, the denominator of the adjustment is smaller than one and $w_P > w_P^0$. The resulting market exposure of the combined portfolio is

$$\beta_{\text{opt}} = w_P \beta_P + w_M = \tfrac{15}{23} \cdot 1.2 + \tfrac{8}{23} \approx 1.13.$$

2.4 (d) Sharpe ratio of the optimal portfolio

The squared Sharpe ratio of the optimal portfolio is made up of the Sharpe ratio of the market and the squared and the squared contribution of the active position, measured as alpha per unit of residual risk

$$S_{\text{opt}}^2 = S_M^2 + \left(\frac{\alpha_P}{\sigma_\varepsilon}\right)^2.$$

Subsituting the respective values yields a Sharpe ratio

$$S_{\text{opt}}^2 = 0.4000^2 + 0.1664^2 = 0.1600 + 0.0277 = 0.1877 \qquad \Longrightarrow \qquad S_{\text{opt}} = \sqrt{0.1877} \approx 0.4332,$$

Alternatively, we can compute the excess return of P_opt and its variance and to find its Sharpe ratio:

$$E[r^e_{\text{opt}}] = \tfrac{15}{23} \cdot 0.15 + \tfrac{8}{23} \cdot 0.10 = 0.1326$$

and

$$\sigma^2_{\text{opt}} = w_P^2 \sigma_P^2 + w_M^2 \sigma_M^2 + 2 w_P w_M \operatorname{Cov}(r_P, r_M)$$

$$\sigma^2_{\text{opt}} = \left(\tfrac{15}{23}\right)^2 (0.1225) + \left(\tfrac{8}{23}\right)^2 (0.0625) + 2\left(\tfrac{15}{23}\right)\left(\tfrac{8}{23}\right)(0.075) \approx 0.0937$$

$$\sigma_{\text{opt}} = \sqrt{0.0937} \approx 0.3061.$$

Calculating the Sharpe ratio now, arrives at the same level as before:

$$S_{\text{opt}} = \frac{E[r^e_{\text{opt}}]}{\sigma_{\text{opt}}} = \frac{0.1326}{0.3061} \approx 0.4332.$$

The optimal portfolio has a Sharpe ratio of 0.4332, which exceeds both $S_P \approx 0.4286$ and $S_M = 0.4000$, confirming that combining the two portfolios dominates holding either one in isolation.